Estimating in Building Construction

Chapter 7 Labor

What is Labor Hour

- One worker working for one hour

Labor Productivity

Productivity rate = Labor hours / Quantity

- Based on historical data
- For average or standard conditions

Labor Productivity

Example

- It took 170 labor hours to complete 10,000 sf of slab.
  - Productivity rate = 170/10,000 = 0.017 labor hour per sf

Labor Productivity

Example 7-1

- Type of work – 8” x8” x 16” Concrete Masonry Units
- Quantity = 1,700 sf
- Labor hours = 170
- Productivity rate = 170/1,700 = 0.10 labor hour per sf

Labor Hours

Labor hours = Quantity takeoff
x Productivity rate

Labor Hours

Adjusted labor hour = Labor hours
x Productivity factor

- Productivity factor:
  - > 1 for below average conditions
  - 1 for average
  - < 1 for above average conditions

Labor Hours

- Productivity factor:
  - When the conditions slow the progress of work, a productivity factor greater than one is used, thus increasing the number of labor hours to complete a unit of work
  - A productivity factor of greater than one is used when conditions are worse than average, and a productivity factor of less than one is used when conditions are better than average.

Factors Affecting Productivity

- Availability of trained craft persons
- Climatic conditions
  - Too hot or cold
  - Rain or snow
  - Wind

Factors Affecting Productivity

- Working conditions
  - Limited working space
  - Limited storage
  - Waiting for materials deliveries
  - Restricted movement around the site

Factors Affecting Productivity

- Other Consideration
  - Breaks: bathroom, coffee, water
  - Clean-up time
  - Availability of equipment

Example 7-2 Labor hours

- Type of work = 8″x8″x16″ Concrete masonry units - decorative
- Quantity of work = 1,000 sf
- Productivity rate = 0.1 labor hours per sf
- Productivity factor = 1.1
- Crew (5 persons) = 3 masons and 2 helpers
- 60% labor hours will be performed by masons and 40% by their helpers
- Labor hours = 1,000 sf x 0.1 labor hours per sf = 100 labor hours
- Adjusted labor hours = 100x1.1 = 110 labor hours
- Masons labor hours (3/5) = 0.60x110 = 66 labor hours
- Mason helper labor hours (2/5) = 0.40x110 labor hours = 44 labor hours

Cycle Time Analysis

$$\text{Productivity rate} = \frac{\text{Average cycle time} \times \text{Productivity factor} \times \text{Crew size}}{\text{System Efficiency} \times \text{Quantity per cycle}}$$

- Used for repetitive cycles
  - Excavation estimating
  - Hauling from pit to job site

Example 7-3 Productivity rate using cycle time

- Type of work – Hauling materials from the borrow pit
- Average cycle time = 35 minutes
- Truck capacity = 17 tons
- Crew = One driver
- Productivity factor = 0.95
- System efficiency = 45 minutes per hour

Productivity rate = (35 x 0.95 x1) / (45x17)

= 0.0435 labor hour per ton

Productivity Rate using Rate of Progress for Linear Tasks

$$\text{Productivity rate} = \frac{\left(\frac{\text{Quantity}}{\text{Rate of progress}} + \text{Travel time}\right) \times \text{Crew size}}{\text{System efficiency} \times \text{Quantity}}$$

- Used for linear tasks
  - Paving or striping a road
  - Placing concrete curbs

Example 7-4 Productivity Rate Using Rate of Progress for Linear Task

- Type of work = Placing concrete curb
- Quantity = 2,200’
- Rate of progress = 3’ per minute
- Crew = One operator and two helpers
- Productivity factor = 0.95
- System efficiency = 45 minutes per hour
- Travel time = 115 minutes

Productivity rate = {(2,200/3 + 115) x 3} / (45 x 2,200)

= 0.0257 labor hours per foot

Published Productivity Rates

Figure 7.1

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Example See Book Page 56

Productivity for an 8”x8”x16” decorative concrete block is 0.118 labor hour per sf

04 22 Concrete Unit Masonry

Productivity Rate

04 22 10 – Concrete Masonry Units

	04 22 10.14 Concrete Block, Back-Up	Crew	Daily Output	Labor-Hours	Unit	2007 Bare Costs Material	Labor	Equipment	Total	Total Incl O&P
1100	6" thick	D-8	430	.093	S.F.	1.98	3.19		5.17	7
1150	8" thick		395	.101		2.15	3.47		5.62	7.65
1200	10" thick		320	.125		2.87	4.29		7.16	9.65
1250	12" thick	D-9	300	.160		3.11	5.35		8.46	11.50
	04 22 10.16 Concrete Block, Bond Beam									
0010	**CONCRETE BLOCK, BOND BEAM**									
0020	Not including grout or reinforcing									
0130	8" high, 8" thick	D-8	565	.071	L.F.	2.19	2.43		4.62	6.10
0150	12" thick	D-9	510	.094		3.05	3.14		6.19	8.15
0525	Lightweight, 6" thick	D-8	592	.068		2.20	2.32		4.52	5.95
0530	8" high, 8" thick	"	575	.070		2.19	2.39		4.58	6.05
0550	12" thick	D-9	520	.092		3.60	3.08		6.68	8.65
2000	Including grout and 2 #5 bars									
2100	Regular block, 8" high, 8" thick	D-8	300	.133	L.F.	4.14	4.57		8.71	11.50
2150	12" thick	D-9	250	.192		5.55	6.40		11.95	15.85
2500	Lightweight, 8" high, 8" thick	D-8	305	.131		4.60	4.50		9.10	11.90
2550	12" thick	D-9	255	.188		6.10	6.30		12.40	16.25
	04 22 10.18 Concrete Block, Column									
0010	**CONCRETE BLOCK, COLUMN**									
0050	Including vertical reinforcing (4-#4 bars) and grout									
0160	1 piece unit, 16" x 16"	D-1	26	.615	V.L.F.	17.50	20.50		38	50.50
0170	2 piece units, 16" x 20"		24	.667		23	22		45	59.50
0180	20" x 20"		22	.727		33.50	24.50		58	74
0190	22" x 24"		18	.889		47.50	29.50		77	97
0200	20" x 32"		14	1.143		51.50	38		89.50	115
	04 22 10.19 Concrete Block, Insulation Inserts									
0010	**CONCRETE BLOCK, INSULATION INSERTS**									
0100	Inserts, styrofoam, plant installed, add to block prices									
0200	8" x 16" units, 6" thick				S.F.	1.13			1.13	1.24
0250	8" thick					1.13			1.13	1.24
0300	10" thick					1.33			1.33	1.46

Unions

- Set wage rules
  - Prevailing hourly wages
  - Fringe benefits
  - Holidays
- How is the union to work with?
  - Prefer to talk or strike

Open Shop

- Employees of the open shop contractor or subcontractor are having greater flexibility with multiple skills.
- Advantage
  - Less restrictive than unions
- Disadvantage
  - No apprentice program for training

- The union carpenter has gone through a structured apprentice program to become a union carpenter

Labor Burden

- Wages paid are known as the bare hourly wage rate / bare wage rate
- Contractor incurs additional costs which are known as labor burden. Examples are:
- **Cash equivalents and allowances**: paid for the employee own tools or for the use of his personal vehicle
- Treated as wages
- Different from reimbursements
  - Based on actual costs

Labor Burden

- **Payroll taxes**: Both the employer and the employee are required to pay 6.2% social security tax. Both have to pay 1.45% Medicare tax.
- Social Security
  - 6.2% on first $106,800 of wages (2010)
- Medicare
  - 1.45% of all wages

Labor Burden

- **Unemployment insurance**: The Federal Unemployment Tax Act (FUTA) and State Unemployment Tax Act (SUTA) at the rate of 6.2% on the first $7,000 of each employee's wage.

Labor Burden

- **Workers' compensation insurance**: It is compulsory insurance for all employees. Medical insurance, lost wages due to injury on the job, occupational illness, killed on the job
- It is paid entirely by the employer and is labor burden.
- Varies by:
  - Worker class (high risk task such as roofing, low risk task such as office work)
  - Company's accident history
- Compensation rates are based on actual losses, the companies must take measures to improve safety and reduce the cost of accidents.
- On average, workers compensation insurance = $7.25 per $100 of wages

Labor Burden

- **General Liability Insurance**: Provides the company with protection from lawsuits arising from negligence of company employees
  - Bodily injury, property damage or loss, damage to one's reputation.
- **Insurance benefits**: Companies often provide employees with health, dental, life or disability insurance

Labor Burden

- **Retirement Contributions**: Employer contributes money to the employee's retirement program. An employer may contribute $0.5 for every $1.0 the employee contributes on 6% of an employee's wages.

Labor Burden

- **Union payments**: When employees belong to a union, the employer is responsible for making payment to the union according to the union contract. Provide for retirement and insurance and apprentice program.
  - Benefits for union employees
  - Training programs

Vacation, Holiday, Sick Leave

- Accounted for by excluding these hour from billable hours

Burdened Hourly Wage Rate

$$\text{Burdened wage rate} = \frac{\text{Wages + Benefits}}{\text{Billable hours}}$$

- Billable hours exclude:
  - Vacation
  - Holidays
  - Sick leave
  - Other unbillable time

Example 7-5 Burdened hourly wage rate

Craft = Mason

Wage rate = $29 per hour

Hours worked = 50 hours per week for 20 weeks and 40 hours per week for 29 weeks

Paid vacation, holidays, and sick leave = 3 weeks at 40 hours per week

Overtime= Time-and-a-half for any hours over 40 per week

Gas allowance = $100 per month

Annual bonus = 500

Social security = 6.2% on the first $106,800 of wage

Medicare = 1.45% of all wage

Example 7-5 Burdened hourly wage rate

Federal Unemployment Tax Act = 0.8% on the first $7,000 of wage

State Unemployment Tax Act = 4.5% on the first $18,000 of wages

Workers compensation insurance = $7.25 per $100 of wages

General liability insurance = 0.75% of wages

Health insurance (company's portion) = $300 per month per employee

Retirement = 0.75 per $1 contributed by the employee on 6% of the employee's wages.

Example 7-5 Burdened hourly wage rate

Wages = (40 hours per week x 52 weeks x $29 per hour)
+ (10 hour per week x 20 weeks x $29 per hour x 1.5)
+ 100 per month x 12 months +500

Wages = $70,720

Social security = o.062 x 70720 = $4384.64

Medicare = 0.0145 x 70720 = $1025

FUTA = 0.008 x 7000 = $56

SUTA = 0.045 x 18000 = $810

Worker compensation insurance = 7.25/100 x 70720
= $5127

General liability insurance = 0.0075 x 70720 = $530

Health insurance = $300/month x 12 = $3600

Example 7-5 Burdened hourly wage rate

Assume that employee take full retirement benefits

Retirement = 0.06 x 70720 x 0.75 = 3182

Benefits = 4385+1025+56+810+5127+530+3600+3182

= $ 18,715

Billable hours = 50 hours per week x 20 weeks + 40 hours per week x 29 weeks = 2160 hour

Burdened hourly wage rate = (70,720 + 18,715)/2160

= $41.41 per hour

Pricing Labor

Labor cost = Adjusted labor hours
x Weighted average burdened wage rate

Example 7-6 Determining labor cost

From example 7-2

There were 110 labor hours

From Figure 7-2

Weighted average burdened wage rate = 35.58

Labor cost = 110 x 35.58

= $3913.8

Quantity	Craft	Bare Hourly Wage Rate	Labor Burden	Burdened Hourly Wage Rate	Total Labor Cost/Hour
3	Mason	29.00	12.41	41.41	124.23
2	Helpers	18.00	8.84	26.84	53.68
5				Total	177.91
				Weighted Average Burdened Wage Rate	35.58

FIGURE 7.2. Weighted Average Burdened Wage Rate.

Standard Crews

Figure 7.3

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Crews

Crew No.	Bare Costs		Incl. Subs O & P		Cost Per Labor-Hour	
Crew C-23A	**Hr.**	**Daily**	**Hr.**	**Daily**	**Bare Costs**	**Incl. O&P**
1 Labor Foreman (outside)	$30.75	$246.00	$47.90	$383.20	$32.39	$49.70
2 Laborers	28.75	460.00	44.75	716.00		
1 Equip. Oper. (crane)	39.80	318.40	60.00	480.00		
1 Equip. Oper. Oiler	33.90	271.20	51.10	408.80		
1 Crawler Crane, 100 Ton		1717.00		1888.70		
3 Conc. buckets, 8 C.Y.		518.40		570.24	55.88	61.47
40 L.H., Daily Totals		$3531.00		$4446.94	$88.28	$111.17
Crew C-24	**Hr.**	**Daily**	**Hr.**	**Daily**	**Bare Costs**	**Incl. O&P**
2 Skilled Worker Foremen	$40.00	$640.00	$62.25	$996.00	$38.17	$59.05
6 Skilled Workers	38.00	1824.00	59.15	2839.20		
1 Equip. Oper. (crane)	39.80	318.40	60.00	480.00		
1 Equip. Oper. Oiler	33.90	271.20	51.10	408.80		
1 Lattice Boom Crane, 150 Ton		1852.00		2037.20	23.15	25.47
80 L.H., Daily Totals		$4905.60		$6761.20	$61.32	$84.52
Crew C-25	**Hr.**	**Daily**	**Hr.**	**Daily**	**Bare Costs**	**Incl. O&P**
2 Rodmen (reinf.)	$41.30	$660.80	$67.75	$1084.00	$32.33	$53.67
2 Rodmen Helpers	23.35	373.60	39.60	633.60		
32 L.H., Daily Totals		$1034.40		$1717.60	$32.33	$53.67

Thank You